Fizika

9-sinf

Laboratoriya ishi №3

Mavzu: Suyuqlikning sirt taranglik koeffitsiyentini aniqlash.

Dars rejasi

Laboratoriya ishini bajarishda texnika xavfsizligi qoidalari bilan tanishish

Laboratoriya ishini bajarishda kerakli bo‘lgan jihozlar bilan tanishish

Laboratoriya ishini tajribada ko‘rsatish

Ishning maqsadi

- Suyuqlikning sirt taranglik koeffitsiyentini aniqlashni o‘rganish.

Kerakli jihozlar

- Sezgir dinamometr, shtativ, uchburchak, to‘rtburchak va aylana shaklidagi similar, suv solinadigan idish, chizg‘ich, shtangensirkul.

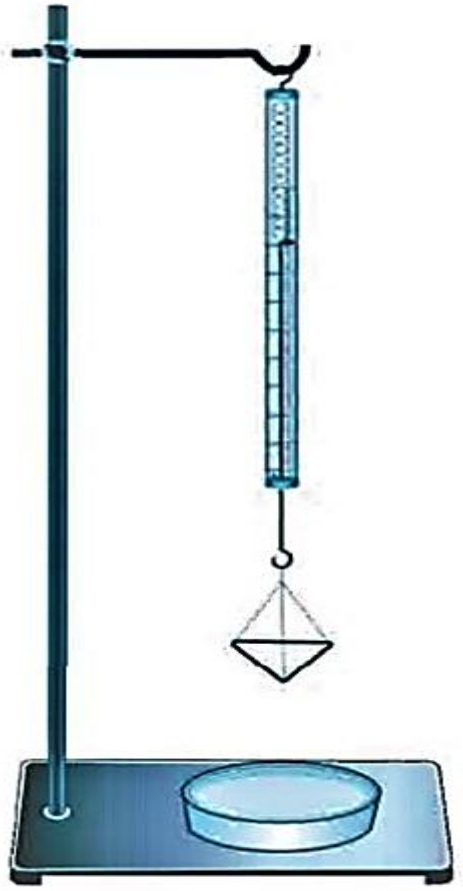
Ishni bajarish tartibi

1. Dinamometrni shtativga o‘rnating.

2. Berilgan shakldagi simning perimetrini o‘lchang.

3. Dinamometrning pastki halqasiga berilgan shakldagi simni iling va og‘irlik kuchi F_1 ni o‘lchang.

4. Idishdagi suvni ko‘tarib, dinamometrga osilgan simga tekkizing.



5. Idishni sekin pastga tortib, simning suvdan uzilishi paytidagi dinamometrning ko‘rsatishini $\boldsymbol{F_2}$ ni yozib oling.

6. $\boldsymbol{F = F_2 - F_1}$ formuladan sirt taranglik kuchini toping.

7. $\boldsymbol{\sigma = \frac{F}{2l}}$ formula yordamida sirt taranglik koeffitsiyentini hisoblang.

8. Tajribani 3-5 marta takrorlab, quyidagi formula yordamida sirt taranglik koeffitsiyentining o‘rtacha qiymatini hisoblang.

$$\boldsymbol{\sigma = \frac{\sigma_1 + \sigma_2 + \sigma_3}{3}}$$

Ishni bajarish tartibi

9. O‘lchash va hisoblash natijalari quyidagi jadvalga yoziladi:

№	m, kg	l, m	σ, N/m	$\sigma_{o'rt}$, N/m
1				
2				
3				

Mustaqil bajarish uchun topshiriqlar

1. Mavzuga oid savollarga javob yozing. (103-bet)

2. Laboratoriya ishi natijalarini tahlil qiling va xulosa yozing.